

Spatio-Temporal Bayesian Hierarchical Modeling for Climate-Driven Financial Risk in Kenya

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1. Introduction

Agricultural market dynamics within Kenya are deeply vulnerable to erratic seasonal precipitation. This paper presents a programmatic approach to estimating the risk profiles of localized food price inflation driven by lagging spatial climate stressors.

2. Methodology & Data Infrastructure

We implement a robust, reproducible multi-stage data architecture: 1. **Ingestion:** Raw historical data is processed via Python into an indexed SQLite database. 2. **Spatial Topology:** A 47x47 Contiguity Weight Matrix is constructed to map boundary coordinates. 3. **Statistical Formulations:** We establish a generalized linear mixed model under a Log-Normal likelihood:

$$Y_{it} \sim \text{Log-Normal}(\mu_{it}, \sigma^2)$$

$$\mu_{it} = \beta_0 + \beta_1 X_{i,t-3} + s_i + \gamma_t$$

Where: * Y_{it} represents the commodity market price in county i during month t . * $X_{i,t-3}$ is the localized historical rainfall aggregate lagged by three months. * s_i represents the spatial random effects across neighboring coordinates. * γ_t accounts for fixed macro-economic temporal trends.

3. Statistical Analysis & Model Summary

Below is the live execution printout compiled directly from our Bayesian simulation chains:

```
[1] "Waiting for local Markov Chain Monte Carlo chains to finalize processing..."
```